

Martin Meinel

POSTDOCTORAL RESEARCHER · MACHINE LEARNING FOR HEALTH

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Profile

Postdoctoral researcher with expertise in leveraging genetic and multi-omic data to inform diagnostics and early-stage target discovery. Developed a diagnostic classifier distinguishing early mycosis fungoides from eczema and psoriasis using bulk, single-cell, and spatial transcriptomic data, now translated into the **patent-pending Erysk-TargetDisk RUO assay**, available through Dermagnostix GmbH. Complementarily, developed methods to identify candidate causal genes and the cellular context in which their regulation is strongest by integrating genetic and single-cell data, advancing precision therapeutic targeting. This work was selected for oral presentation at the NeurIPS 2025 AI4D3 workshop and was supported by the Chan Zuckerberg Initiative.

Research Experience

Developed gene-expression based diagnostic classifier for early mycosis fungoides

Helmholtz Munich

PHD STUDENT

Oct 2021 - Aug 2026

- Identified 6-gene biomarker panel discriminating early mycosis fungoides from eczema and psoriasis from bulk RNA-seq data using SAFARI, a computational framework developed for robust biomarker discovery.
- Reduced the biomarker panel to a 2-gene classifier with the option to defer clinically ambiguous cases, achieving **91% sensitivity and 94% specificity** in the qPCR development cohort.
- Validated classifier generalizability across **8 independent international cohorts**.
- Showed earlier detection than standard-of-care methods (histopathology and clinical presentation) in four longitudinally followed patients.
- Characterized marker expression in single-cell and spatial transcriptomics (10x Visium) data by integrating public datasets, revealing *LCK* expression in T cells (MF) and *HOMER1* in keratinocytes (eczema, psoriasis).

Causal inference for early-stage target discovery (FocusMR)

Helmholtz Munich, CZI Residency

PHD STUDENT

Oct 2023 - Aug 2026

- Identified candidate causal disease genes with the cell states where regulation is strongest using FocusMR, an attention-based single-cell Mendelian Randomization framework.
- Prioritized 34 candidate causal eczema genes at cell-type level resolution with **four loci mapping to fine-grained sub-cell-states** by integrating eczema GWAS and single-cell genetics data (OneK1K).
- Demonstrated biological plausibility of the discovered cell states by recovering known disease-associated biology, including *PRKCQ-AS1* in naive/central-memory CD8⁺ T cells enriched for PD-1 signaling and CD28 costimulation.
- Resolved a *TAPBPL*-defined CD4⁺ T cell state that **spans several annotated subtypes** rather than localizing to one, enriched for a shared Th17/Th22 program.
- Developed a Bayesian strategy to assess robustness of candidate cell states across independent individuals, supporting context-aware target prioritization.

Publications & Manuscripts

- Meinel, M.[†], Langreder, N.[†], et al. (2026). LCK and HOMER1 gene-expression based classifier distinguishes early mycosis fungoides from eczema and psoriasis. Manuscript in 3rd-round revision at *Science Translational Medicine*. preprint: <https://www.medrxiv.org/content/10.1101/2025.10.18.2533802>.
- Garger, D.[†], Meinel, M.[†], Dietl, T., Hillig, C., Garzorz-Stark, N., Eyerich, K., de Angelis, M., Eyerich, S., Menden M.P. (2023). The impact of the cardiovascular component and somatic mutations on ageing. *Aging Cell*, <https://doi.org/10.1111/accel.13957>.
- Fischer, F., Doll, A., Uereyener, D., Roenneberg, S., Hillig, C., Weber, L., Hackert, V., Meinel, M., Farnoud, A., Seiringer, P., et al. (2023). Gene expression based molecular test as diagnostic aid for the differential diagnosis of psoriasis and eczema in formalin fixed and paraffin embedded tissue, microbiopsies and tape strips. *The Journal of Investigative Dermatology*, S0022-202X(23)00156-2. <https://doi.org/10.1016/j.jid.2023.02.015>.
- Farnoud, A., Ohnmacht, A. J., Meinel, M., & Menden, M. P. (2022). Can artificial intelligence accelerate preclinical drug discovery and precision medicine?. *Expert opinion on drug discovery*, 17(7), 661–665. <https://doi.org/10.1080/17460441.2022.2090540>

[†] symbolizes shared first authorship.

Skills

Genomic Technologies

Single-Cell RNA-seq analysis & integration (scanpy, Harmony, scVI, scANVI, scCODA)
Spatial transcriptomics analysis & integration (scanpy, scANVI, 10x Visium)
Statistical Genetics (Mendelian Randomization, GWAS, HEIDI, Colocalization, PLINK)
Bulk RNA-sequencing analysis (DESeq2, edgeR, fgsea, clusterProfiler)

Programming & Data Analysis

Python (NumPy, Pandas, scikit-learn, PyTorch), R, Bash

High-Performance & Cloud Computing

Hydra, Slurm, Docker, Git, AWS

Languages

English (Fluent), German (Native)

Education

Postdoc with focus on context-dependent eQTL mapping

HELMHOLTZ CENTER MUNICH | CASALE LAB

Munich, Germany

Aug 2026 - present

PhD Candidate in computational biology and statistical genetics

HELMHOLTZ CENTER MUNICH | LMU MUNICH | CHAN ZUCKERBERG INITIATIVE RESIDENCY

Munich, Germany

Oct 2021 - Aug 2026

- Supervised by Dr. Michael Menden, Dr. Natalie Garzorz-Stark & Dr. Francesco Paolo Casale

M. Sc. in Data Engineering & Analytics

TECHNICAL UNIVERSITY OF MUNICH

Munich, Germany

Oct 2018 - Nov 2020

B. Sc. in Applied Computer Science

UNIVERSITY OF BAYREUTH

Bayreuth, Germany

Oct 2014 - Oct 2017

Supervision, Teaching & Review

- 2025 Supervisor of three students for their master's theses and internships.
- 2024 Teaching assistant for Computational Modeling for System Genetics.
- 2025 Reviewer for the **RECOMB** Conference.
- 2024 Reviewer for the **ML for Health** workshop.

Munich, Germany

TU Munich

Seoul, KOR

Vancouver, CA

Honors & Awards

Patent

- 2025 Published patent application WO/2025/051923, "Differential Diagnosis of Mycosis Fungoides: Method and System", leveraging SAFARI to discover biomarkers distinguishing early MF from eczema & psoriasis.

Munich, Germany

Oral presentations

- 2025 Oral presentation at **NeurIPS AI4D3 workshop** on how to identify cellular contexts of causal genes enhancing early-stage drug target discovery.
- 2025 Oral presentation at **EORTC CLTG** about the identification and clinical translation of biomarkers in early MF.

San Diego, USA

Athens, Greece

Chan Zuckerberg Initiative Residency

- 2024 Selected for CZI residency for developing cell-context-aware target discovery methods to identify cellular contexts of causal genes for precise therapeutic targeting.

Redwood, USA

Invited speaker session

- 2023 Presented "Biocomputational Analysis of Big Data" to researchers in dermatology at the EADV-ESDR Summer Research Workshop: Big Data and Translation in Dermatology.

Freiburg, Germany

Poster presentations

- 2025 Highlight poster with oral presentation on cell-state-aware single-cell Mendelian Randomization at **RECOMB Genetics**.
- 2023 Presented the SAFARI framework for rare disease biomarker discovery at the **ML for Health** workshop.
- 2025 Presented work on cell-state-aware single-cell Mendelian Randomization at the **Helmholtz AI conference**.
- 2025 Presented classifier development and validation results for early MF diagnosis from psoriasis and eczema at **DKTK meeting**.

Seoul, KOR

New Orleans, USA

Karlsruhe, Germany

Freiburg, Germany

Working experience

Opitz Consulting GmbH

SOFTWARE ENGINEER WORKING STUDENT

Munich, Germany

Oct 2019 - Oct 2020

- Built a Docker skeleton for Apache Zeppelin, enabling users to start a docker container running Apache Zeppelin on OpenShift with one click.
- Created a web application for Allianz IDS GmbH, allowing employees to track, assign, and assess investment orders within the department.

REHAU Industries SE & Co. KG

SOFTWARE ENGINEER WORKING STUDENT

Bayreuth, Germany

Nov 2017 - Mar 2019

- Developed an ALEXA skill for the 2018 FRONTALE window exhibition, demonstrating several features of the smart window.
- Implemented a web app for Medi Bayreuth to track player performance and fatigue for injury prevention.